

Ques. WHAT IS AN LTPS LCD? WHAT ARE ITS ADVANTAGES AND DISADVANTAGES?

A. Samiuddhin

Ans. Low-temperature polycrystalline silicon (or LTPS) LCD—also called LTPS TFT LCD—is a new-generation technology product derived from polycrystalline silicon materials. Polycrystalline silicon is synthesised at relatively low temperatures ($\sim 650^{\circ}\text{C}$ and lower) as compared to traditional methods (above 900°C).

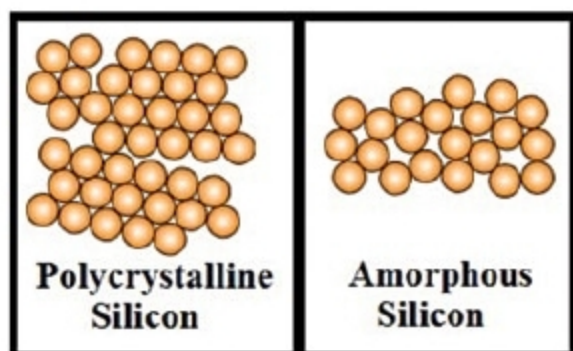


Fig. 1: Amorphous silicon versus polycrystalline silicon (Credit: Wikipedia)

Standard LCDs found in many consumer electronics, including cell-phones, use amorphous silicon as the liquid for the display unit. Recent technology has replaced this with polycrystalline silicon, which has boosted the screen resolution and response time of devices.

Row/column driver electronics are integrated onto the glass substrate. The number of components in an LTPS LCD module can be reduced by 40 per cent, while the connection part can be reduced by 95 per cent. The LTPS display screen is better in terms of energy consumption and durability, too.

LTPS LCDs are increasingly becoming popular these days. These have a high potential for large-scale production of electronic devices such as flat-panel LCD displays or image sensors.

Amorphous silicon lacks crystal structure, whereas polycrystalline silicon consists of various crystallites or grains, each having an organised lattice (Fig. 1).

Advantages of an LTPS LCD display are:

- Dynamic and rich colours
- Fast response and less reflective
- High picture resolution

Some of its disadvantages are:

- Deteriorates faster than other LCDs
- High cost

Q2. WHAT IS ARTIFICIAL INTELLIGENCE (AI), AND WHAT ARE ITS VARIOUS APPLICATIONS? PLEASE GIVE SOME DIY EXAMPLES BASED ON ARDUINO, NODEMCU AND RASPBERRY PI.

Pamarthi Kanakaraja

A2. AI emphasises on making intelligent machines work and react like human beings, including speech recognition, planning, problem solving and learning. It involves programming intel-

ligence displayed by humans and other animals.”

AI finds application in robotics, virtual assistants, the Internet, finance and economics, exploration and research, healthcare, education, automotive, video games, defence, businesses and on mobile devices. For more on its applications, please refer to ‘The Latest In AI and Its Applications’ article published in January issue.

There are many AI-based DIY projects using Arduino, NodeMCU and Raspberry Pi available on the Internet.

A simple example is ‘Artificial Intelligence-Based Chatbot For Appliance Control’ DIY article published in November 2018 issue of EFY. It uses NodeMCU and Arduino IDE along with Chatfuel platform to create an AI chatbot. Wiring diagram of the project is shown in Fig. 2.

This AI-chatbot is used to control

electrical appliances using voice commands from Facebook Messenger. First, add AI rules in Chatfuel by writing all permissible words or phrases a user may use—such as ‘Turn on the light’ or ‘Switch on the light’—for interaction. After the system is connected to the Internet, open the chatbot from Facebook Messenger installed on your cellphone and provide it with instructions using the microphone, to turn

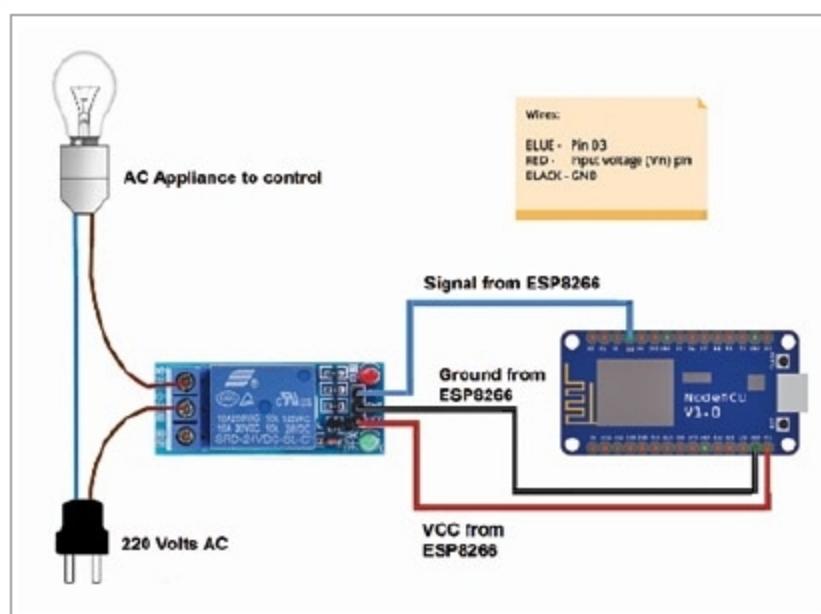


Fig. 2: Wiring diagram of AI-based chatbot

ligence or making computers behave like humans. AI-powered devices try to imitate the natural intelligence of humans, mimicking the cognitive functions that humans use to perform tasks.

There are many definitions of AI and one such definition given on Wikipedia is, “In computer science, AI—which is sometimes called machine intelligence—is intelligence demonstrated by machines, in contrast to the natural

the light off or on.

AI-based projects with Raspberry Pi can be found on websites such as <https://rpi.ai.com> and www.artificial-intelligence.blog

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